

Business Requirements Document (BRD)

Project Name: E-Commerce Portfolio Platform Version: 1.0 Date: [Insert Date] Author: [Your Name/Team]

1. Executive Summary

Project Overview

The **E-Commerce Portfolio Platform** is a **full-stack web application** designed to enable businesses to manage their online storefronts, including product catalogs, user accounts, orders, invoicing, and geographic location management. This platform supports **administrative, staff, and customer roles** with distinct permissions and workflows.

Business Objectives

- Provide a **scalable, modular e-commerce backend** for managing products, orders, and user interactions.
- Enable **role-based access control** (admin, staff, customer) for secure data management.
- Support **real-time inventory tracking** and **order processing** with automated invoicing.
- Facilitate **geographic location management** (states, cities) for shipping and logistics.
- Ensure **scalability and maintainability** through a well-structured codebase and modular architecture.

Expected Outcomes

- A **fully functional e-commerce backend** integrated with a **React-based frontend**.
- Automated email notifications** for order confirmations, invoices, and password resets.
- Role-based dashboards** for admins, staff, and customers.
- Seamless API integration** for future extensions (e.g., payment gateways, CRM systems).

2. Project Scope

In-Scope Features

| **Feature Area** | **Description** | |-----|-----| | **User Management** | Registration, login, password reset, role-based access (admin, staff, customer). | | **Product Catalog** | CRUD operations for products (name, description, price, images, category). | | **Order Processing** | Order creation, status tracking (pending, completed, cancelled), inventory updates. | | **Invoicing** | Invoice generation, payment status tracking, and deletion. | | **Geographic Management** | State and city registration for shipping and logistics. | | **Shopping Cart** | Add/remove items, clear cart, and calculate total price. | | **Email Notifications** | Automated emails for order confirmations, invoices, and password resets. | | **Admin Dashboard** | Analytics, user management, product inventory, and order tracking. | | **API Integration** | RESTful API for frontend communication (Next.js). |

Out-of-Scope Items

- **Multi-currency support** (single currency assumed).
- **Advanced analytics** (e.g., sales trends, customer behavior).
- **Mobile app development** (web-focused only).
- **Third-party payment gateways** (e.g., Stripe, PayPal) integration.
- **Multi-language support** (English-only).

Key Assumptions

- **MongoDB** will be used for data storage (NoSQL flexibility).
- **JWT (JSON Web Tokens)** will handle authentication.
- **React (Next.js)** will be the primary frontend framework.
- **Express.js** will serve as the backend API.
- **Nodemailer** will handle email sending (SMTP-based).
- **Soft deletes** will be used for data integrity (e.g., users, products, orders).

3. Business Requirements

3.1 Functional Requirements

User Management

Requirement	Description
Registration	Allow new users to sign up with email, password, and contact details.
User Login/Logout	Secure authentication with JWT tokens.
Password Reset	Email-based password recovery flow.
Role-Based Access	Admins, staff, and customers have distinct permissions.
User Profile Management	Users can update their personal details (e.g., phone, addresses).

Product Management

Requirement	Description
Product CRUD	Create, read, update, and delete products with categories.
Product Search	Filter products by name, category, or price range.
Inventory Tracking	Real-time stock updates when orders are placed.
Product Images	Support for multiple product images (e.g., Cloudinary integration).

Order Processing

Requirement	Description
Order Creation	Users can add products to cart and checkout.
Order Status Tracking	Orders can be marked as pending, completed, or cancelled.
Inventory Deduction	Automatically reduce stock when an order is placed.
Order History	Users can view their past orders.

Invoicing

Requirement	Description
Invoice Generation	Automatically generate invoices for completed orders.
Payment Status	Track whether an invoice is paid or unpaid.
Invoice Deletion	Admins can delete unpaid invoices.

Geographic Management

Requirement	Description
State & City Registration	Admins can add states and cities for shipping addresses.
Address Management	Users can associate addresses with their accounts.

Shopping Cart

Requirement	Description
Add/Remove Items	Users can modify cart contents before checkout.
Cart Persistence	Cart persists across sessions for logged-in users.
Total Price Calculation	Automatically calculate subtotal, tax, and shipping.

Email Notifications

Requirement	Description
Order Confirmation	Email sent after order placement.
Invoice Delivery	Email sent with invoice details.
Password Reset	Email sent with reset link.

Admin Dashboard

Requirement	Description
User Management	Admins can view, edit, and delete users.
Order Tracking	Admins can view all orders and their statuses.
Product Inventory	Admins can manage product stock levels.
Analytics Dashboard	Basic sales and user activity metrics.

3.2 Non-Functional Requirements

Category	Requirement
Performance	API responses should load within < 2s under normal load.
Scalability	System should handle 10,000+ concurrent users without degradation.
Security	- JWT tokens with 1-hour expiry. - Passwords stored as hashed BCrypt (256-bit). - CORS and CSRF protection enabled. - Rate limiting to prevent brute-force attacks.
Availability	99.9% uptime (SLA).
Data Integrity	Soft deletes for users, products, and orders.
Backup & Recovery	Daily automated backups of MongoDB data.
Compliance	GDPR-compliant data handling (user data deletion requests).

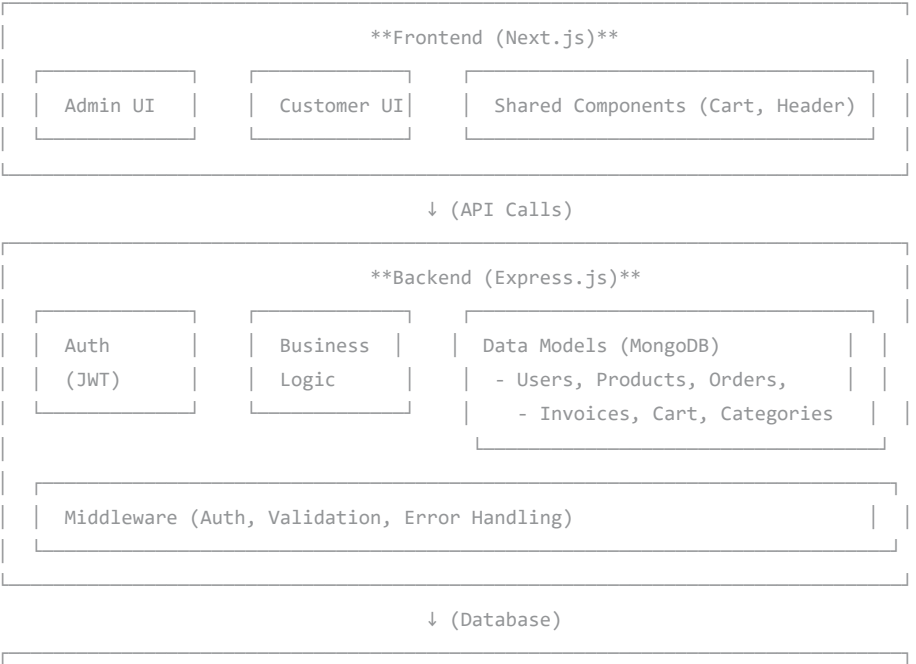
3.3 User Stories

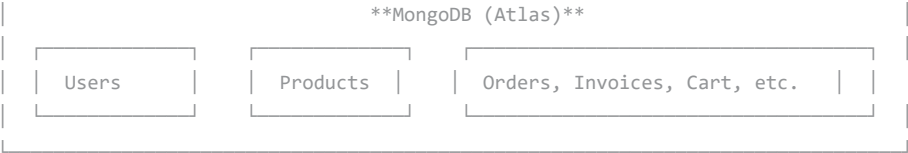
User Type	Story	Acceptance Criteria
Customer	As a customer, I want to browse products so I can find what I need.	

Product list page loads with search/filter functionality. | | | As a customer, I want to add items to my cart so I can purchase them later. | - "Add to Cart" button works; cart persists across sessions. | | | As a customer, I want to checkout with my payment details so I can complete my purchase. | - Checkout flow validates payment info and generates an order. | | | As a customer, I want to receive an order confirmation email so I know my purchase was successful. | - Email sent with order details after checkout. | | **Admin** | As an admin, I want to manage products so I can update inventory and pricing. | - CRUD operations for products; stock levels update automatically on orders. | | | As an admin, I want to view all orders so I can track sales and resolve issues. | - Admin dashboard shows order history with filters. | | | As an admin, I want to generate invoices so I can bill customers. | - Invoice generation with payment status tracking. | | **Staff** | As staff, I want to process orders so I can fulfill customer requests. | - Staff can update order status (e.g., "shipped"). | | | As staff, I want to manage user accounts so I can support customers. | - Staff can view/edit user details (excluding sensitive data). |

4. Technical Architecture Overview

4.1 High-Level System Architecture





4.2 Technology Stack

| **Layer** | **Technology** | **Purpose** | |-----|-----|-----|-----|
-----| | **Frontend** | Next.js (React) | SPAs with server-side rendering for SEO and performance. | | **Backend** | Express.js | RESTful API for frontend communication. | | **Database** | MongoDB (Atlas) | NoSQL database for flexible schema and scalability. | | **Authentication** | JWT (JSON Web Tokens) | Secure user authentication and role-based access control. | | **Email** | Nodemailer | SMTP-based email sending for notifications. | | **Validation** | Zod | Runtime type checking and validation for API inputs. | | **Testing** | Jest, Supertest | Unit and integration testing for backend APIs. | | **DevOps** | Docker, GitHub Actions | Containerization and CI/CD pipelines. |

4.3 Integration Points

- **Third-Party Services:**
 - **Cloudinary** (for product image storage).
 - **SMTP Provider** (e.g., SendGrid, Mailgun) for email delivery.
- **Future Extensions:**
 - **Payment Gateways** (Stripe, PayPal).
 - **CRM Integration** (e.g., HubSpot, Salesforce).
 - **Multi-Language Support** (i18n libraries).

5. User Personas & Use Cases

5.1 Target Users

- Admin updates stock levels.

3. **Alternative Flows:**

- Admin filters products by category/price.
- Admin exports product data to CSV.

4. **Postcondition:** Product inventory is updated.

Use Case 3: Staff Processes an Order

1. **Precondition:** Staff is logged in.

2. **Main Flow:**

- Staff views "Orders" dashboard.
- Staff updates order status (e.g., "shipped").
- Staff generates an invoice if not already done.

3. **Alternative Flows:**

- Staff cancels an order due to stock issues.
- Staff resolves a customer dispute.

4. **Postcondition:** Order status is updated; customer notified.

6. Success Criteria

6.1 Key Performance Indicators (KPIs)

| Metric | Target | |-----|-----| | **API Response Time** | < 2s for 95% of requests. | | **System Uptime** | 99.9% availability. | | **User Registration** | 90% of users complete registration without errors. | | **Order Completion Rate** | 85% of carts are converted to orders. | | **Email Delivery Rate** | 95% of emails are successfully delivered. |

6.2 Acceptance Criteria

Feature	Acceptance Criteria	User Registration
	Users can register with valid email and password. - Passwords are hashed before storage.	
	- Product images are stored and displayed.	Product Management - Admins can CRUD products with categories.
	Order Processing - Users can add items to cart and checkout. - Inventory is deducted on order completion.	
	Invoicing - Invoices are auto-generated for completed orders. - Admins can mark invoices as paid/unpaid.	Email Notifications - Emails are sent for order confirmations and password resets.
	Role-Based Access - Admins can access all features; staff/customers have restricted permissions.	Geographic Management - Admins can register states and cities for shipping.

6.3 Business Value Metrics

Metric	Business Impact	Revenue Growth
	Increased sales through seamless checkout and product management.	Customer Satisfaction Higher retention via order tracking and support features.
	Operational Efficiency Reduced manual work for order processing and inventory management.	Scalability Ability to handle growth without performance degradation.
	Cost Savings Automated invoicing and email notifications reduce administrative overhead.	

7. Implementation Timeline

7.1 High-Level Milestones

Phase	Duration	Deliverables
Discovery & Planning	2 weeks	BRD, user personas, technical architecture.
Backend Development	6 weeks	API endpoints, models, authentication, and core business logic.
Frontend Development	6 weeks	Next.js UI, cart, checkout, and admin dashboards.
Integration	3 weeks	Connect frontend to backend; test email and payment flows.
Testing	2 weeks	Unit, integration, and UAT testing.
Deployment	1 week	Deploy to production; monitor performance.
Go-Live	Ongoing	Post-launch support and iterative improvements.

7.2 Dependencies

- MongoDB Atlas Account (for database hosting).

- **SMTP Credentials** (for email sending).
- **Cloudinary Account** (for image storage).
- **Frontend Team** (for UI development).

7.3 Risk Considerations

Risk	Mitigation Strategy	Database Performance Issues	Security Vulnerabilities	User Adoption	Scalability Bottlenecks
		Use indexing for frequently queried fields (e.g., <code>userId</code> in <code>orders</code>).	Regular security audits; use HTTPS and rate limiting.	Implement retry logic for email and payment services.	Conduct training sessions for admins/staff.
					Monitor API response times; optimize queries with pagination.

8. Appendix

8.1 Glossary

Term	Definition	JWT	CRUD	RESTful API	MongoDB Atlas
		JSON Web Token: A secure way to authenticate users via tokens.	Create, Read, Update, Delete: Basic operations for managing data.	A standard for designing networked applications using HTTP requests.	A cloud database service for MongoDB.

8.2 References

- Technical Docs:
 - [Express.js Documentation](#)
 - [MongoDB Atlas Guide](#)
 - [Next.js API Routes](#)
- Standards:

- GDPR Compliance: General Data Protection Regulation
- Security Best Practices: OWASP Top 10

Approval: | Role | Name | Signature | Date | |-----|-----|-----|-----| | Business Owner | [Name] | | [Date] | | Tech Lead | [Name] | | [Date] |

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